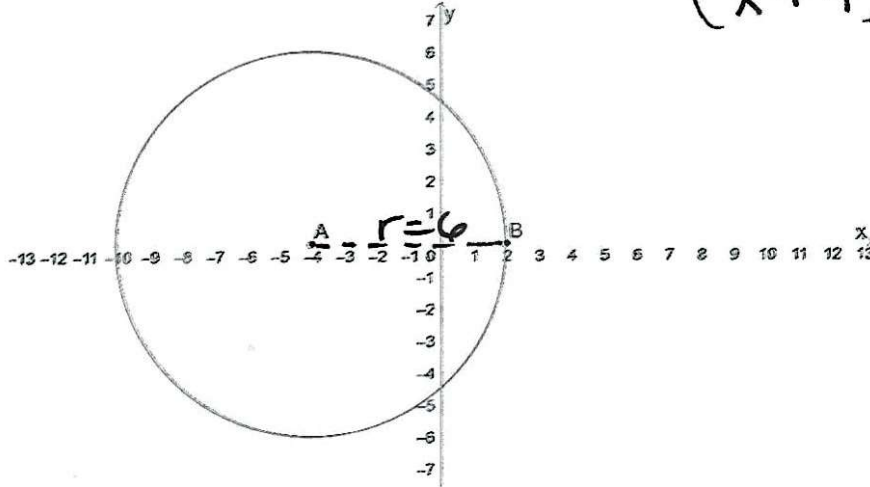


Unit 12: Worked Examples

1. Circle A is shown with center $A(-4,0)$. Point $B(2,0)$ lies on the circle. Write the equation that represents circle A.

$$(x+4)^2 + y^2 = 36$$



2. Rewrite $x^2 + y^2 - 12x - 14y = -81$ in center-radius form. Then, identify the center, radius, domain, and range of the circle.

$$\begin{aligned} x^2 - 12x + y^2 - 14y &= -81 \\ (x^2 - 12x + 36) + (y^2 - 14y + 49) &= -81 + 36 + 49 \\ (x-6)^2 + (y-7)^2 &= 4 \end{aligned}$$